

SiPM-Wall Bars: Top/Bottom Combination & Vertical Position

n_TOF EAR2 X17 — run 224404, wall-plastic MIP coincidences

Dylan Neff

analysis and slides prepared with Claude (Anthropic AI assistant)

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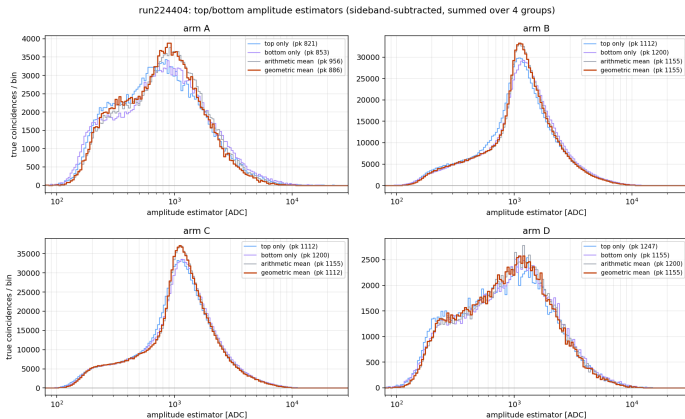
Setup: what the two ends give us

- ▶ Wall bars are **vertical**, ~ 50 cm, read by **top** and **bottom** SiPMs (detn odd/even; 4 groups of 4 bars per arm).
- ▶ Two ends of the same deposit $\Rightarrow$ three things to try:
 1. combine amplitudes for a position-independent MIP estimate (geometric vs arithmetic mean);
 2. vertical position from amplitude asymmetry $\ln(A_{\text{top}}/A_{\text{bot}})$;
 3. vertical position from timing $t_{\text{bot}} - t_{\text{top}}$.
- ▶ Same clean MIP sample as before: same-arm wall $\times$ plastic coincidence, late tof, sideband-subtracted (16/100).

$$\begin{aligned}A_{\text{top}} &= A_0 e^{-(L/2-y)/\lambda} \\A_{\text{bot}} &= A_0 e^{-(L/2+y)/\lambda}\end{aligned}$$

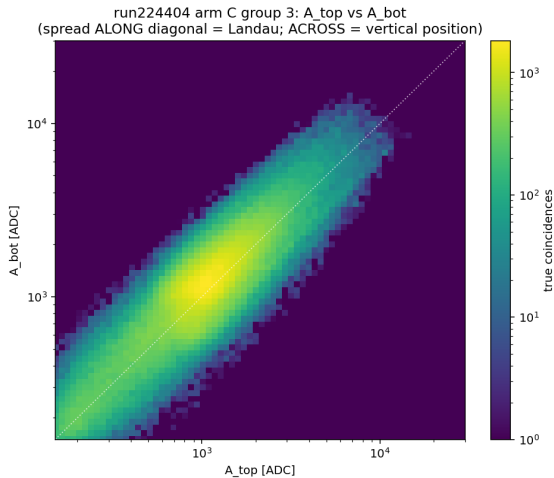
$$\begin{aligned}\sqrt{A_{\text{top}}A_{\text{bot}}} &= A_0 e^{-L/2\lambda} \quad (\text{no } y) \\ \ln \frac{A_{\text{top}}}{A_{\text{bot}}} &= \frac{2y}{\lambda}, \quad t_{\text{bot}} - t_{\text{top}} = \frac{2y}{v_{\text{eff}}}\end{aligned}$$

(2) Combine the ends: geometric mean is (slightly) best



B/C: geometric mean gives the sharpest MIP peak ($\text{FWHM/peak} \approx 0.90\text{--}0.96$ vs $1.0\text{--}1.1$ single-ended); arithmetic mean nearly ties it. The two means agree because attenuation is *weak* ($\lambda \gg L$, next slides). A/D: no MIP peak (accidental-dominated).

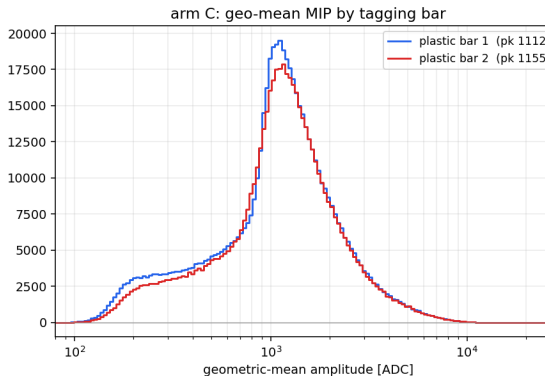
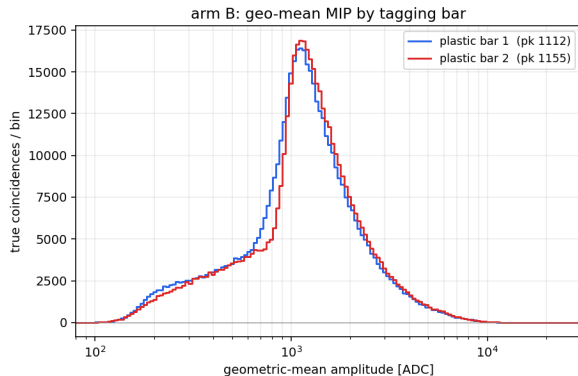
Why geo $\approx$ arith here: A_{top} vs A_{bot}



- Spread **along** the diagonal = the (large) Landau energy fluctuation, *correlated* between the two ends.
- Spread **across** the diagonal = vertical position (attenuation asymmetry) — much smaller.
- Geometric mean projects along the diagonal $\Rightarrow$ keeps the energy, cancels position. Since the across-spread is small, arithmetic mean does almost as well.

(1) Split by plastic bar: MIP peak is bar-independent

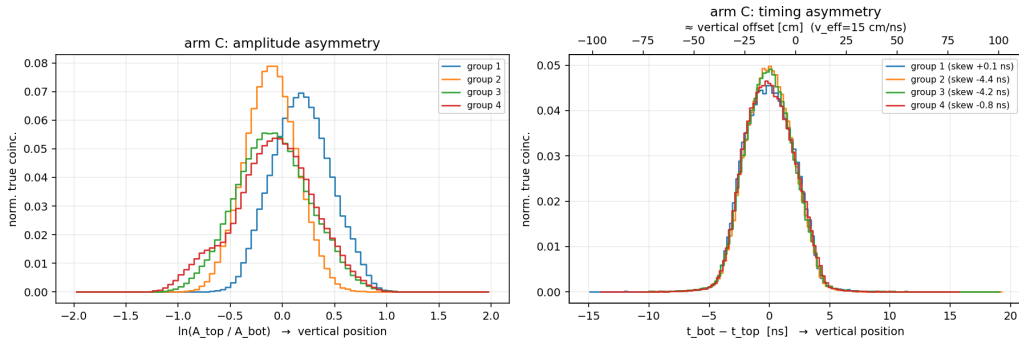
run224404: wall MIP spectrum split by which plastic bar tagged it



Wall geo-mean MIP spectrum split by which plastic bar tagged the coincidence. Peaks agree to one bin (1112 vs 1155 ADC) on both arms $\Rightarrow$ merging the two bars (as the main analysis does) is safe; the low-amp shoulders differ, tracking the bar $\leftrightarrow$ u-half geometry.

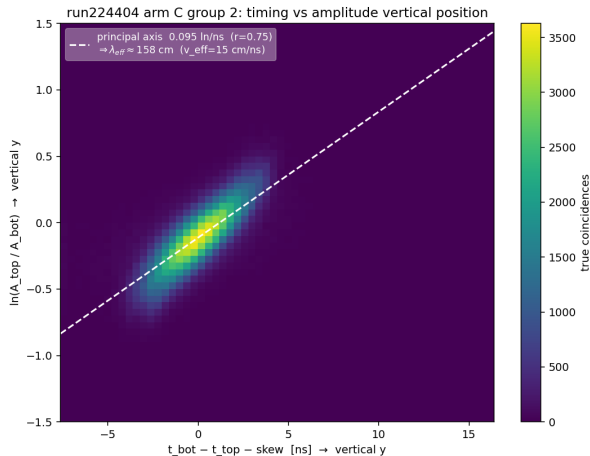
(3) Vertical position: amplitude and timing both work

run224404: vertical-position observables (sideband-subtracted, arm C)



Sideband-subtracted, per group (arm C). Both $\ln(A_{\text{top}}/A_{\text{bot}})$ and the cable-skew-corrected $t_{\text{bot}} - t_{\text{top}}$ span the bar; the top axis converts timing to cm ($v_{\text{eff}} = 15 \text{ cm/ns}$).

The two estimators agree $\Rightarrow$ real vertical coordinate



- ▶ Tight correlation ridge ($r \approx 0.75$): timing and amplitude track the *same* vertical y .
- ▶ Slope = $v_{\text{eff}}/\lambda_{\text{eff}} \Rightarrow \lambda_{\text{eff}} \approx 100\text{--}160$ cm (consistent across all 8 B/C groups) — i.e. $\sim 2\text{--}3\times$ the bar length, hence weak attenuation.
- ▶ Single-hit vertical resolution $\sigma_y \approx \mathbf{10\text{--}13}$ cm from *either* method ($\sim 4\text{--}5$ elements over 50 cm); combining them is tighter still.

Summary

- ▶ **Plastic bars:** plotting the wall MIP spectrum separately per tagging bar gives the *same* peak $\Rightarrow$ the standard bar-merged spectrum is unbiased.
- ▶ **Top/bottom amplitude:** use the **geometric mean** $\sqrt{A_{\text{top}}A_{\text{bot}}}$ — sharpest, position-independent MIP peak; arithmetic mean is a close second because $\lambda_{\text{eff}} \gg L$.
- ▶ **Vertical position is recoverable**, and cross-validated two ways: amplitude asymmetry and top/bottom timing agree ($r \approx 0.75$), both giving $\sigma_y \approx 10\text{--}13$ cm over the 50 cm bar.
- ▶ Byproduct: $\lambda_{\text{eff}} \approx 100\text{--}160$ cm and $v_{\text{eff}} \approx 15$ cm/ns effective — can be pinned exactly with a triggered/known-position sample.
- ▶ Caveat: 4 bars are ganged per group $\Rightarrow$ this is a per-group vertical y , with the horizontal (u) coordinate only at group / plastic-bar granularity.